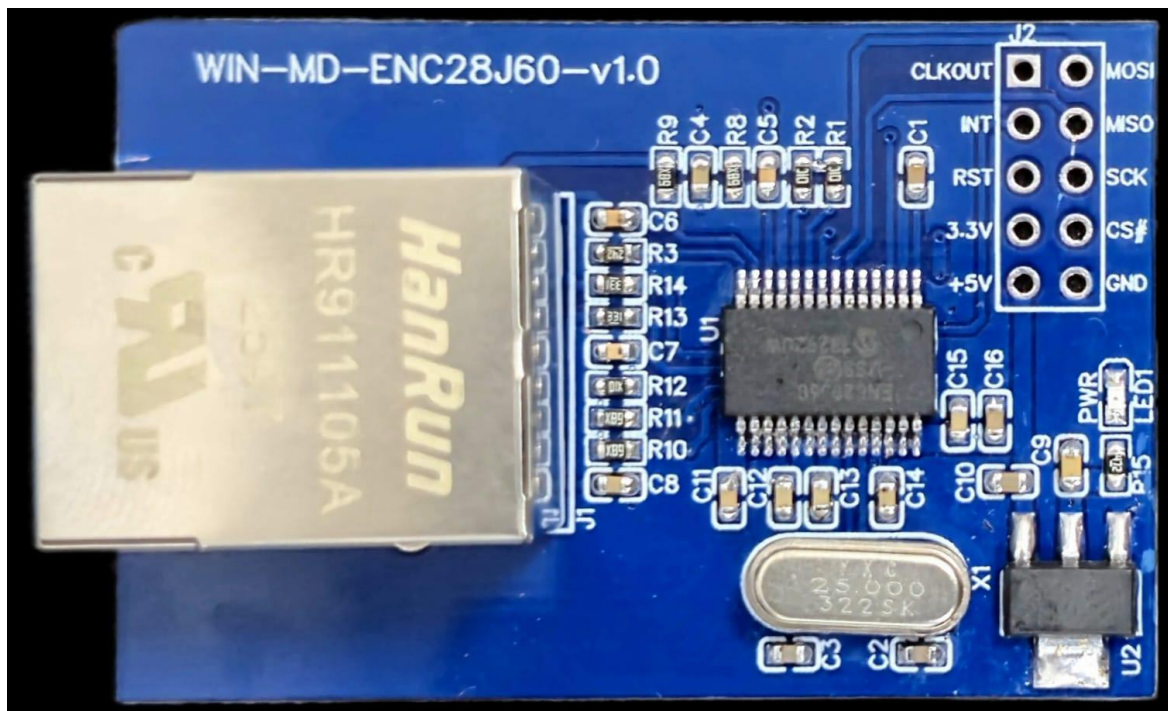


Technical Datasheet ENC28J60-Based Ethernet Module

The Ethernet module is based on the ENC28J60, a standalone Ethernet controller with an integrated MAC and PHY. It interfaces with the host microcontroller via a high-speed SPI bus and provides 10 Mbps Ethernet connectivity for embedded applications.

ENC28J60 Module: -



Unlike hardwired TCP/IP controllers, the ENC28J60 operates at the Ethernet frame level and requires the host microcontroller to implement the TCP/IP protocol stack (such as LwIP or uIP). This offers greater flexibility and control over network communication but increases firmware complexity.

The module includes an onboard RJ45 connector with integrated magnetics, ensuring reliable signal integrity and simplified hardware design. It is well-suited for cost-sensitive and customizable networking applications in embedded and industrial systems.

The ENC28J60 module is ideal for applications where full control over networking protocols is required, such as custom communication stacks, lightweight web servers, and specialized industrial communication interfaces.

Specifications

General –

Interface	SPI (Serial Peripheral Interface)
Ethernet	10 Mbps, RJ45 with Integrated Magnetics
I/O Connector	2 × 5 Pin Header, 2.54 mm Pitch
Dimensions	32.7 mm (L) × 50.0 mm (W)
Power Supply	5V DC / 3.3V DC Input

Certifications

Additional Features: -

Based on ENC28J60

Integrated MAC and PHY supporting 10BASE-T Ethernet

SPI interface for communication with host MCU

Requires external TCP/IP stack implementation (LwIP/uIP)

Onboard RJ45 connector with integrated magnetics

Suitable for low-cost and customizable embedded networking applications

Compact design for easy integration into industrial and IoT systems

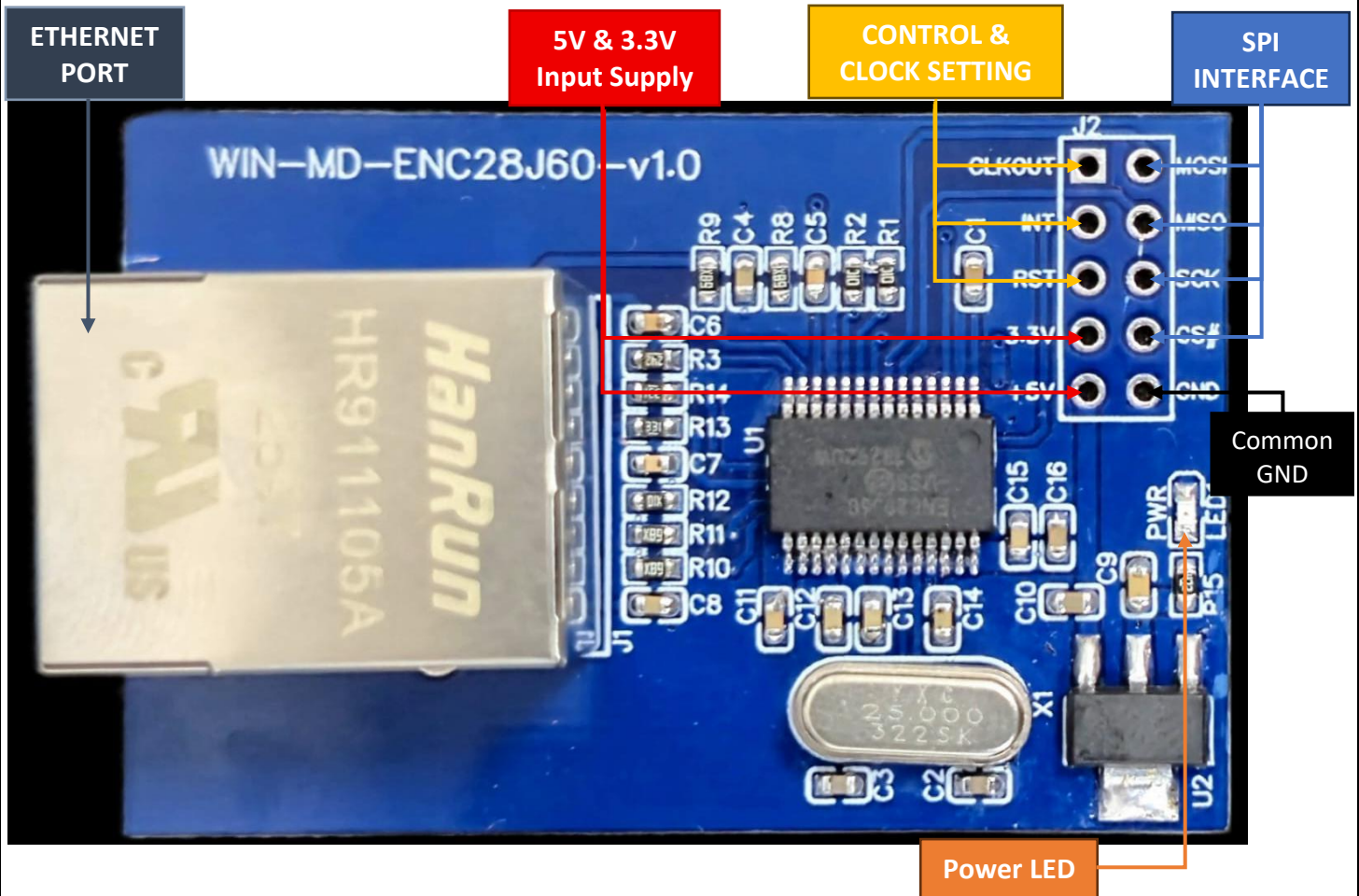
Note: -

Use standard CAT5e or higher Ethernet cable for reliable communication.

Ensure proper grounding and stable power supply for optimal performance.

SPI lines should be kept short and properly routed to minimize signal noise.

Adequate firmware stack (TCP/IP) is required on the host controller.



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